

Applied Physics for EC/EE Stream-2022 Scheme

Module-1(Study any 5 Questions)

1. Explain de-Broglie Hypothesis, derive an expression for de-Broglie Wavelength ‘
2. Setup time independent schrodinger’s wave equation
3. State and explain the Heisenberg Uncertainty principle with relevant equations, Based on the principle, prove that free electrons do not exist inside the nucleus.
4. What is wave function? Give its physical significance and properties
5. Discuss the wave functions, probability densities and energy level for a particle in a box by considering the ground and the first two excited State

OR

What are the properties of a wave function? Give the qualitative explanation of Max-Born’s interpretation of wave function

6. What are matter waves, Mention their properties, Define phase and Group Velocity
7. Obtain the expression for Eigenvalue and Eigenfunction for particles in a box.

Module-2(Study any 5 Questions)

1. Mention any three assumptions of quantum free electron theory? Discuss the dependence of Fermi factor on temperature and Consequent effect on probability of occupation of energy levels
2. State the main assumptions of QFET, Give an account of Fermi Velocity and Temperature
3. Derive Clausius Mossotti equation
4. What is polarization? Describe the different polarization mechanism.
5. Explain the dependence of resistance on temperature of a superconductor? Explain the different types of Superconductors
6. Write a short note on (i)Superconductors (ii)Critical Magnetic Field (iii)Critical Temperature (iv)Squids
7. Discuss the solid, liquid and gaseous dielectrics with one example each. Explain the role of dielectrics as an electrical insulator.

Module-3(Study any 5 questions)

1. Write a short note on (i)Spontaneous and stimulated emission (ii)Requisites and conditions of Laser (iii)How matter interacts with radiation
2. Define the following terms (i)Population Inversion (ii)Meta stable state (iii)Acceptance angle

- (iv) Numerical aperture (v) Fractional index change
3. Derive an expression for numerical aperture of an optical fiber with a suitable diagram
 4. Explain the various types of optical fibers with proper diagrams, also describe how P-P communication is enabled through optical fibers
 5. What is meant by attenuation and attenuation coefficient, Mention the expression for attenuation coefficient in a fiber of length L , Mention advantages and disadvantages
 6. Derive the expression for energy density of radiation using Einstein's coefficients, Compare the expression with Planck's Equation

Module-4 (Study any 4 questions)

1. Derive wave equation in terms of Electric field using Maxwell's equation
OR
Derive wave equation for electromagnetic waves in vacuum in terms of electric field using Maxwell's equation
2. Explain the terms gradient of a scalar, divergence and curl of a vector. Derive Gauss divergence theorem and write an expression for Stokes theorem
3. Explain Faraday's Laws of Electromagnetic induction, Ampere's Law and express the same in point form
4. Derive the expression for displacement current
5. Elucidate the Transverse nature of EM Waves through Linear Polarization

Module- 5

1. What is the Hall effect? obtain an expression for the Hall coefficient, Define Hole concentration and Electron Concentration. Derive the relation between Fermi Energy and Energy Gap in an intrinsic semiconductor
2. Derive an expression for Electrical conductivity in extrinsic and intrinsic semiconductors
3. Write a short note on photodiode- discuss the importance of power responsivity
Or
With construction and working explain about Phototransistor
4. Explain in detail about the Four-Probe Method, use resistivity as a factor to explain the concept along with its application
5. Describe the construction and working of a semiconductor laser with an energy level diagram.